MATH 11 — ELEMENTARY STATISTICS

§6.1

Continuous Probability Distributions

From “probability of a value” to “area under a curve”

Today's Goal

After completing this section, you should be able to:

- Understand the difference between discrete and continuous probability distributions
- Interpret a continuous probability density function
- Calculate probabilities as areas under a probability density function
- Apply the properties of continuous probability distributions

Learning Target 9

LT.9 Uniform Probability Distributions

I understand what a uniform probability distribution is. I can use a given uniform probability distribution to calculate the probability of an appropriate event.

The Probability of One Value

A continuous random variable X can take any value from 0 to 8.

What is the probability that X is exactly 5?

The bridge from discrete to continuous is the shift from "probability of a value" to "area under a curve".

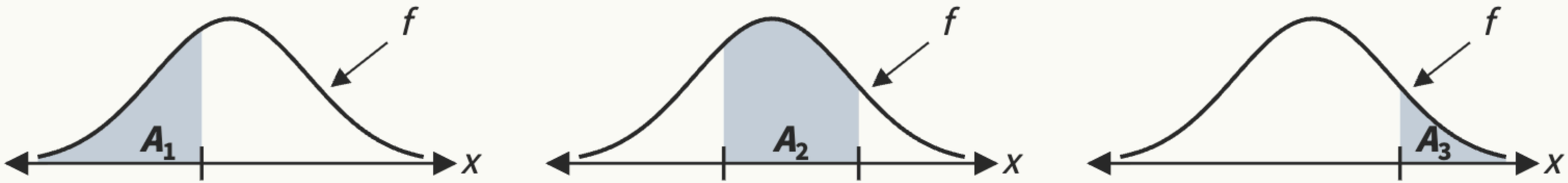
Probability Density Function

Definition 6.1.1

Given a continuous random variable X, the **probability density function** for X is a function f for which the area between its graph and the x-axis over a range of x-values is the probability associated with that range of x-values.

Area Under a Curve

In general, there are three ranges of x-values with which we work.



$$A_1 = P(X \leq a) \qquad A_2 = P(a \leq X \leq b) \qquad A_3 = P(X \geq b)$$

In each, the probability is the shaded area between the density curve f and the x-axis.

Properties

Below are properties of probability density functions, f, and their associated probability distribution functions, P:

1. The entire area under the graph of $f(x)$ and above the x-axis is 1.
2. $P(X = c) = 0$ for any value of c. In other words, the probability that X takes on any specific individual value is zero. This is because the area below the curve, above the x-axis, and between $x = c$ and $x = c$ has no width.

Properties, continued

3. When finding the probabilities associated with a range of x-values (such as in Figure 6.1.2), the inequalities $\leq$ and $\geq$ may be replaced with $<$ and $>$ without affecting the probability. For example, $P(X \leq c) = P(X < c)$.

Is $P(X \geq 3)$ the same as $P(X > 3)$?

Yes. $P(X = 3) = 0$, so the two probabilities are the same.

Continuous PDF

Definition 6.1.3

A **continuous probability distribution function** (continuous PDF) is a probability distribution function for a continuous random variable X that assigns probabilities to intervals of x-values using the corresponding probability density function $f(x)$. The probability assigned to an interval is the area between the probability density function and the x-axis over that interval.

Find Each Probability

Consider the probability density function

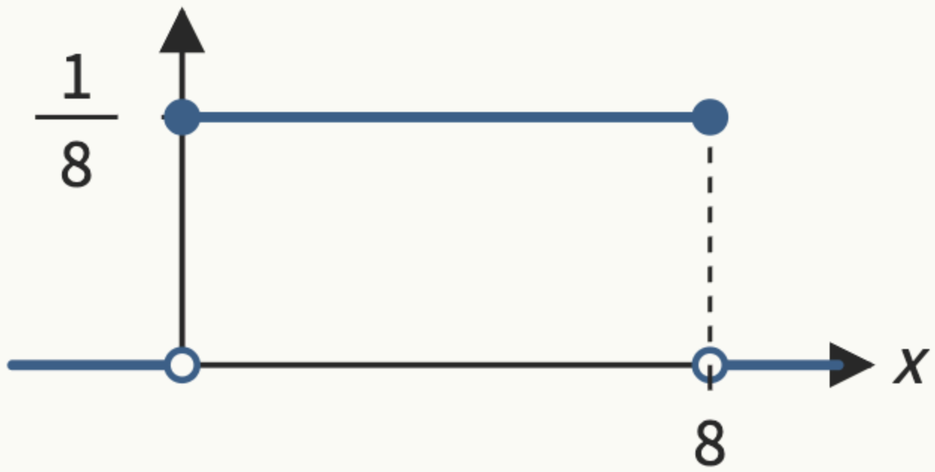
$$f(x) = \begin{cases} \frac{1}{8} & \text{if } 0 \le x \le 8 \\ 0 & \text{otherwise,} \end{cases}$$

graphed here. Find each of the following probabilities.

(a) $P(0 < X < 5)$

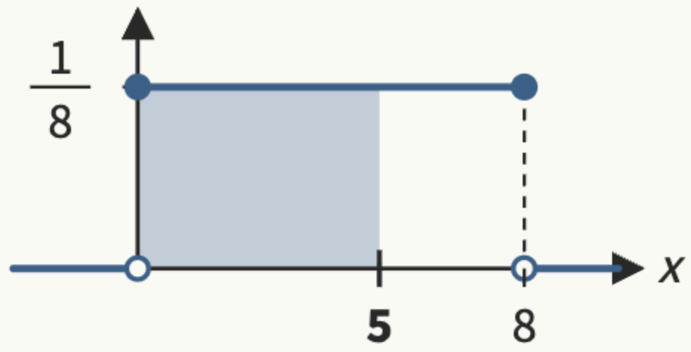
(b) $P(2 < X < 5)$

(c) $P(X > 6)$



Each Probability Is an Area

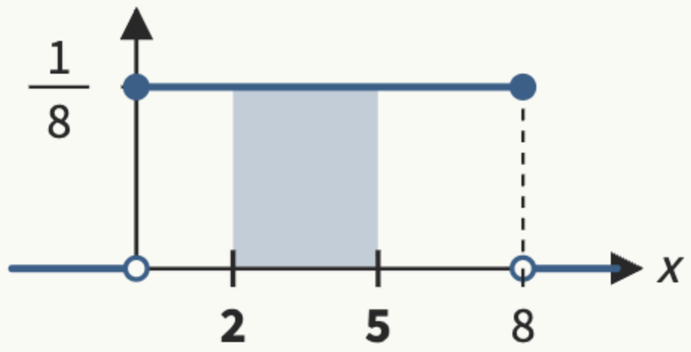
(a) $P(0 < X < 5)$



The rectangle from $x = 0$ to $x = 5$ has width 5, so

$$P(0 < X < 5) = 5 \cdot \tfrac{1}{8} = \tfrac{5}{8}.$$

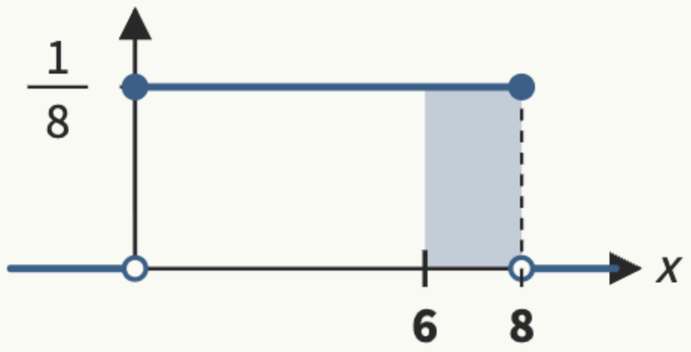
(b) $P(2 < X < 5)$



The rectangle from $x = 2$ to $x = 5$ has width 3, so

$$P(2 < X < 5) = 3 \cdot \tfrac{1}{8} = \tfrac{3}{8}.$$

(c) $P(X > 6)$

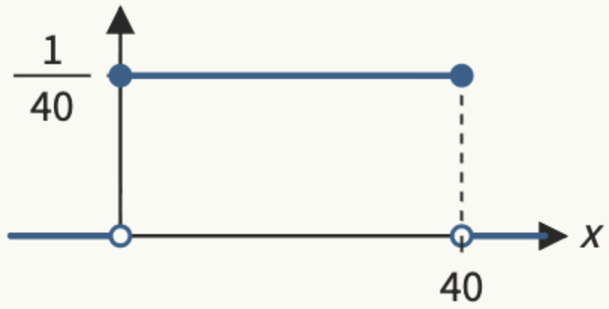


The rectangle from $x = 6$ to $x = 8$ has width 2, so

$$P(X > 6) = 2 \cdot \tfrac{1}{8} = \tfrac{2}{8}.$$

Now You Try: Probability as Area

You Try 6.1.1. A continuous random variable X has the probability density function f graphed below: $f(x) = \frac{1}{40}$ for $0 \leq x \leq 40$, and $f(x) = 0$ for every other value of x.



Find each of the following probabilities. Round each one to **four** decimal places.

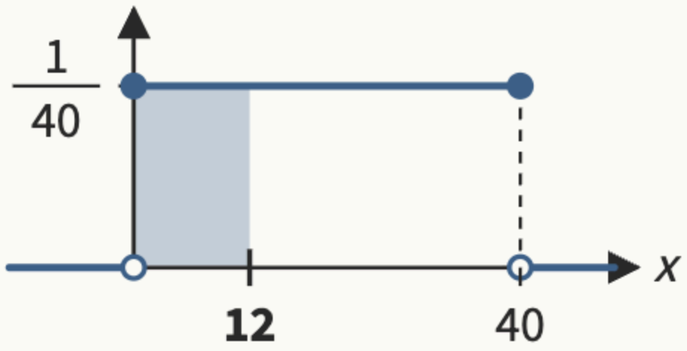
(a) $P(X < 12)$

(b) $P(18 \leq X \leq 27)$

(c) $P(X \geq 32)$

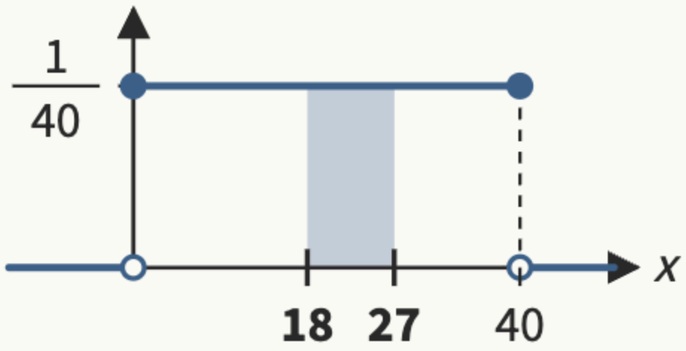
You Try 6.1.1: Checking Your Answer

(a) $P(X < 12)$



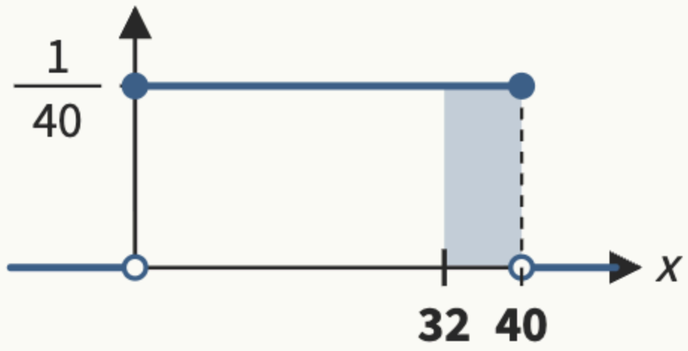
$$P(X < 12)$$
$$= 12 \cdot \tfrac{1}{40} = \tfrac{12}{40}$$
$$= 0.3000$$

(b) $P(18 \leq X \leq 27)$



$$P(18 \leq X \leq 27)$$
$$= 9 \cdot \tfrac{1}{40} = \tfrac{9}{40}$$
$$= 0.2250$$

(c) $P(X \geq 32)$



$$P(X \geq 32)$$
$$= 8 \cdot \tfrac{1}{40} = \tfrac{8}{40}$$
$$= 0.2000$$